

Quantitative analysis of tribological performance in polyamide 6/nano-zeolite composites for prosthetic and orthotic applications: Integration of computer vision, numerical modeling, and experimental evaluation

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ABSTRACT

This study investigated the effect of nano-zeolite (NZ) integration on the tribological properties of polyamide 6 (PA6) nanocomposites using a hybrid approach combining image segmentation, experimental techniques, and numerical modeling. Automated SEM image analysis with U-Net segmentation quantified nanoparticle distribution, while pin-on-disc tests assessed wear performance. A support vector regression (SVR) model linked nanoparticle dispersion, load, and hardness to friction and wear rates. A combined dispersion index (CDI) further quantified nanoparticle distribution. Results demonstrated significant wear rate reductions (41 % at 60 N and 54 % at 100 N) with 5 wt% NZ. The SVR model highlighted the importance of nanoparticle dispersion and load on performance, with hardness having minimal influence. Morphological analysis confirmed smoother worn surfaces with NZ inclusion.

1. Introduction

Polyamide 6 (PA6), composed of recurring amide-6 units, is a polymer renowned for its exceptional physical and chemical properties, making it a versatile material across various industries. Key advantages of PA6 include its high heat resistance, excellent shock and impact resistance, superior mechanical strength, and strong insulation and dielectric properties [1]. Furthermore, the low specific weight of PA6 compared to metals renders it particularly well-suited for lightweight applications.

PA6 is extensively utilized in the medical field for various applications, including the manufacture of surgical sutures, where its biocompatibility and excellent tensile strength contribute to effective wound closure and healing [2]. It is also employed in orthopedic implants and bone tissue engineering due to its durability and resistance to bodily

fluids [3]. Specifically, PA6 has emerged as a valuable material for gradient implants and solid-state drug delivery systems, owing to its osseointegration, absorptive, and biodegradable performance [4]. The primary advantage of these products lies in their remarkable osseointegration capabilities. Once implanted into bone, they facilitate the growth of bone tissue through the material, ultimately leading to complete integration with the body, thereby eliminating the need for surgical removal. The ability to fine-tune gradient properties enables a closer approximation of natural bone structure, while the material free volume allows for the storage and controlled release of drugs via desorption and osseointegration processes. Moreover, the osseointegration rate is influenced by the structure of the polymer-based drug delivery container or implant, with factors such as temperature and pressure during solidification playing a crucial role in the final product performance. However, the widespread use of pure PA6 in load-bearing

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applications is limited by its mechanical weaknesses, low heat resistance, and high friction coefficient [5]. To address these shortcomings, reinforcing PA6 with nanofillers offers a promising avenue to enhance its properties and broaden its range of applications.

Nanocomposite materials represent a significant technological advancement due to the integration of nano-fillers, which offer substantial improvements over traditional micro-reinforcements in polymeric materials, aligning well with industrial and technological objectives. The incorporation of nano-sized fillers enables the production of lighter, stronger, and more cost-effective structures. This enhanced load-bearing capacity is attributed to the exceptional surface contact between the nano-fillers and the host matrix, which can be up to 1000 times greater than that of micro-fillers, particularly in the case of nanofibers [6]. The use of minute quantities of nanoscale inorganic fillers, rather than larger micron-sized particles, effectively enhances the wear, thermal, and mechanical properties of polymers. Optimal improvements in these characteristics are achieved when the nanoparticles are uniformly dispersed within the polymer matrix [7]. Additionally, the introduction of filler nanoparticles into semi-crystalline polymers increases crystallinity during solidification, resulting in smaller crystal grains and a more uniform crystal structure, thereby enhancing the overall uniformity of the polymer matrix [8–10].

In literature, various nanoscale fillers, such as carbon nanotubes, layered silicates, and isotropic nanoparticles like silica, calcium carbonate, and zeolite, were utilized to enhance the mechanical and tribological properties of PA6. For example, Srinath et al. [11] demonstrated that incorporating nano-clay significantly improves the tensile strength and wear resistance of PA6. In another study, they examined the wear properties of nano-clay-reinforced PA6 under both dry and wet conditions using the pin-on-disk test method. Their results showed that while nano-clay enhances wear resistance, the presence of absorbed water diminishes this effect. Additionally, the inclusion of clay, with or without a compatibilizer, led to a notable decrease in both elongation at break and melt flow index [12]. Mohd et al. [13] observed that adding nano-clay to a PA6/PP blend substantially reduces impact resistance. The nano-clay layers restrict the movement of polymer chains, reducing their flexibility during deformation, which increases the material brittleness. Stress accumulation at the sites of silicate layers within the polymer matrix may also contribute to cracking [13]. Furthermore, the anisotropic nature of nano-clay introduces anisotropy into the mechanical properties and shrinkage behavior of the polymer matrix.

The influence of nano-zeolite (NZ) on the mechanical properties of polymer matrices has been extensively studied by several researchers [14–17]. For instance, incorporating up to 5 % NZ into polyurethane was shown to enhance both tensile strength and Young's modulus [17]. Zeping et al. [18] investigated the impact of NZ on the mechanical properties and nucleation of polypropylene, finding that the addition of 3 wt% NZ increases tensile and flexural strength, as well as the nucleation rate. Similarly, the inclusion of NZ in polyethylene has been reported to improve fracture toughness [15]. However, there is a notable lack of research on the effect of NZ incorporation on the wear properties of PA6, particularly for its use as a filtration material in the water treatment industry.

A detailed analysis of nanoparticle dispersion characteristics using computer vision has significantly enhanced the understanding of their effects on different properties. However, the automatic segmentation of nanoparticles remains challenging due to variations in size, the presence of artifacts, noise, and blurred regions in the images. Recent advancements in deep learning have shown great promise in overcoming these challenges in polymer and nanoparticle research [19,20]. Particularly, the U-Net architecture, recognized for its precision in semantic segmentation tasks, offers substantial benefits in accurately differentiating individual nanoparticles, even in cases of overlap. Similar techniques were successfully employed in other applications, such as the automated segmentation of nanoparticles in BF TEM images [21], further demonstrating the efficacy of U-Net binarization for nanoparticle analysis.

High wear resistance is essential in prosthetic and orthotic applications due to the demanding conditions these devices endure in daily use. Prosthetic limbs and orthotic supports are exposed to continuous mechanical stress and friction, which can cause material degradation over time. By enhancing wear resistance, the longevity and reliability of these devices are significantly improved, reducing the need for frequent replacements or repairs. As a result, materials with superior wear resistance not only boost the durability and performance of prosthetic and orthotic devices but also enhance overall comfort and functionality for users.

The integration of NZ into PA6 can potentially enhance the wear resistance of materials used in prosthetic and orthotic applications. NZ, characterized by its high surface area, exceptional hardness, and thermal stability, serves as an effective reinforcing filler in the polymer matrix. When properly dispersed within PA6, NZ not only improves the load-bearing capacity but also reduces friction—an essential factor in wear-intensive settings like prostheses and orthoses. Moreover, the porous nature of NZ promotes more effective stress distribution across the material, reducing the risk of surface wear and extending the lifespan of the device. By combining increased mechanical strength with heightened wear resistance, this NZ-PA6 composite can be well-suited to endure the repeated stresses of daily use, contributing to the durability and overall performance of these critical medical devices.

In this study, the tribological performance of PA6 nanocomposites with tunable tribological properties was evaluated using a combined numerical-visual-experimental approach to assess its suitability for prosthetic and orthotic applications. The polymer was reinforced with 2.5, 5, and 7.5 wt% NZ to enhance wear resistance, lubrication, and impact resistance. The nanocomposite samples were fabricated using a melt processing method. A pin-on-disk test was conducted to evaluate the tribological performance of the nanocomposites. Additionally, a novel automated scanning electron microscopy (SEM) image analysis technique, employing a U-Net-based binarization method, was introduced to measure particle area distribution. Mathematical modeling was utilized to quantify nanoparticle dispersion and to correlate particle distribution, loading intensity, and surface hardness with the friction coefficient and specific wear rate. Ultimately, SEM micrographs of the worn surfaces were analyzed to examine the dominant wear mechanisms at the microscale level. By integrating computer vision, numerical modeling, and experimental testing, this approach improves the accuracy and efficiency of characterizing the microstructural features of nanocomposites, providing precise correlations between morphological characteristics and tribological properties.

2. Material and methods

2.1. Materials and sample preparation

Polyamide 6 (PA6, F223-D) from Akulon (DSM, Netherlands) with a density of 1130 kg/m³ was used as the base material. The nanophase component was nano-zeolite (NZ, Z4A-005-R), sourced from Tsuruta-Cho, Japan, with an average particle size of 50 nm. Both PA6 and nano-zeolite were dried at 80°C for 8 hours prior to mechanical mixing. The compound was then processed using a twin-screw extruder with a 25 mm screw diameter and a 40:1 length-to-diameter ratio. The extruded material was cut into granules using a cutting machine. The extruder operated at a rotational speed of 250 rpm, with heater temperatures set to 220°C, 240°C, 260°C, 270°C, 280°C, and 260°C, progressively from the feed zone to the extrusion mold. Specific formulation details for each compound are listed in Table 1.

2.2. Microstructural analysis

The distribution of nanostructured phases within nanocomposite systems and the primary wear mechanisms were examined using a Field Emission Scanning Electron Microscope (FESEM, IRA3/TESCAN, Czech

Table 1

Composition and formulation of the compounds used in this study.

Sample code	Zeolite (wt%)
PA6	0
PA6/2.5NZ	2.5
PA6/5NZ	5
PA6/7.5NZ	7.5

Republic) with an acceleration voltage of 15 kV. Prior to SEM analysis, both impact-fractured and worn surfaces were coated with gold.

2.3. Hardness test

The hardness measurements were performed using a Shore D hardness tester. The samples were prepared following ASTM D 2240 standards, with dimensions of 35 mm in length, 15 mm in width, and 3 mm in thickness.

2.4. Wear characterization

The tribological properties of PA6 and nanocomposite samples were assessed using the pin-on-disk test in accordance with ASTM G99 standards. For this evaluation, cylindrical samples with a diameter of 8 mm and a length of 30 mm were used. The wear disc was made from AISI 314 steel and had a diameter of 160 mm, a thickness of 10 mm, and a surface roughness of 0.6 μm . Tests were performed under applied loads of 60 N and 100 N, at a constant speed of 0.5 m/s, and with a sliding distance of 6000 m, in an environment with an average temperature and humidity of $35 \pm 5\%$. Sample weights were measured before and after testing using a digital balance with an accuracy of 0.1 mg. Wear was determined by comparing the mass of the samples before and after the tests.

2.5. Image segmentation using U-Net

2.5.1. Dataset preparation

The SEM images of PA6-based nanocomposites, with varying loadings of NZ, served as the primary data for this study. To ensure the dataset quality and relevance, the images were carefully curated by excluding those with very low resolution, significant artifacts, or unclear nanoparticle distribution. The remaining images were then divided into training and testing subsets, with samples randomly selected to represent a range of nano-zeolite dispersions.

2.5.2. Network architecture

The U-Net architecture, introduced by Ronneberger et al. [22], is specifically designed for image segmentation tasks. As illustrated in Fig. 1, the architecture consists of multiple convolutional encoder and decoder layers, culminating in a pixel-wise classification layer. Each encoder layer performs repeated 3×3 convolution operations followed by rectified linear unit (ReLU) activation. These layers then reduce the spatial dimensions of the feature maps through 2×2 max pooling with a stride of 2. To counteract the loss of spatial information during down-sampling, the encoder feature maps are up-sampled and merged with corresponding feature maps from the decoder. These merged maps are then processed by ReLU activations within the decoder. The final classification is carried out via a 1×1 convolutional layer, which maps each feature vector to the appropriate class labels. A sigmoid activation function is applied in the output layer to ensure the predicted pixel values fall within the desired range [23,24].

Since the SEM images are grayscale and have a resolution of 512×512 pixels, the input to the U-Net architecture is formatted as $512 \times 512 \times 1$. This format indicates that each image has a height and width of 512 pixels, with a single channel representing grayscale intensity. The segmented output image, which also uses two shades of gray, retains the same shape of $512 \times 512 \times 1$. This consistent input and output dimensionality allows the network to process and segment the images effectively, without altering their inherent structure.

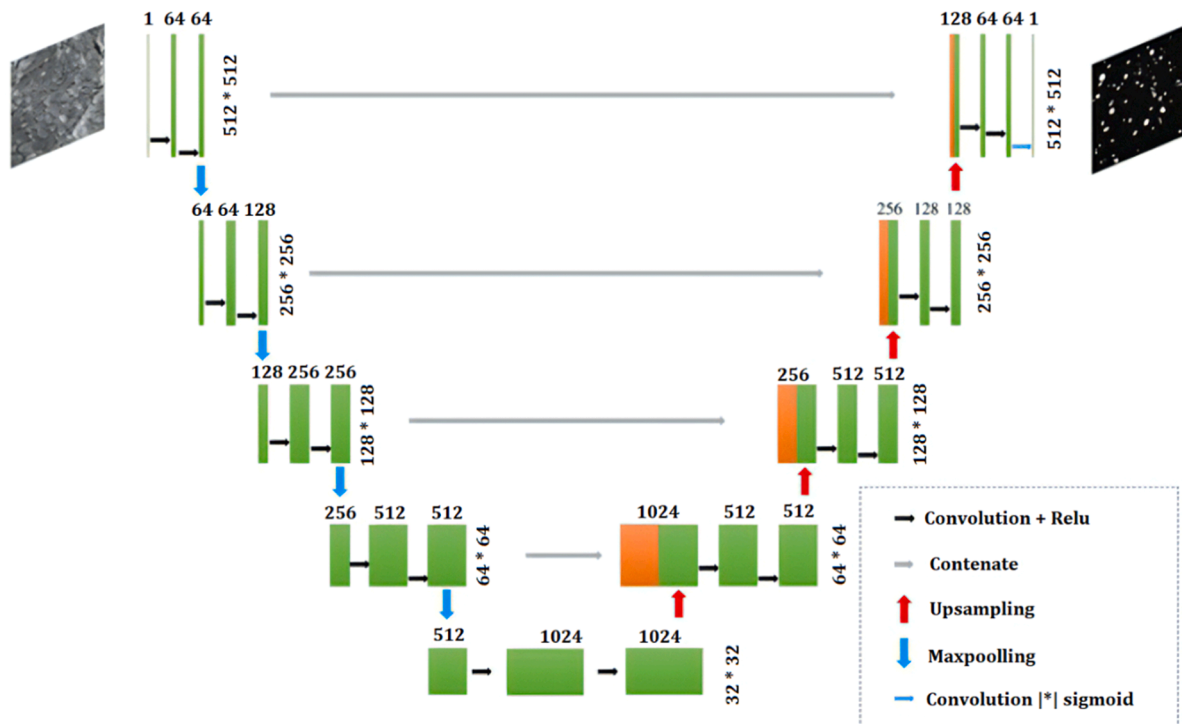


Fig. 1. U-Net architecture for 512×512 input data. Green boxes represent multi-channel feature maps, and orange boxes denote copies from the encoder section.

2.5.3. Model training

The proposed U-Net-based segmentation model was trained and evaluated on an NVIDIA GeForce GTX 1060 GPU using TensorFlow (version 2.8). The network optimization was performed with the Adam optimizer, utilizing a learning rate of $1e-4$ with exponential decay. The batch size was set to 8. As illustrated in Fig. 2, the training process spanned 150 epochs, demonstrating convergence by the end of the training period.

2.5.4. Evaluation metrics

In line with many studies in semantic segmentation, this research employs the mean intersection-over-union (*mIoU*) metric to evaluate pixel-wise accuracy [22,24]. The *mIoU* is calculated as follows:

$$mIoU = \frac{\sum_i^N IoU_i}{N} \quad (1)$$

$$IoU = \frac{area(predicted) \cap area(label)}{area(predicted) \cup area(label)} \quad (2)$$

In these equations, "area" represents the number of pixels, where "predicted" refers to the pixels identified as nanoparticles, and "label" refers to the ground truth pixels corresponding to nanoparticles. The index i denotes the class indices, and N is the total number of classes. A higher *mIoU* value, approaching one, indicates better performance.

To evaluate object-level performance, this study also uses three additional metrics: precision, recall, and the F_1 score. These metrics are defined as follows:

$$F_1 score = \frac{2Precision \times Recall}{Precision + Recall} \quad (3)$$

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

3. Results and discussions

3.1. Performance evaluation of the trained U-Net model

Based on the present test dataset, the trained U-Net model achieved a precision of 84.19 %, a recall of 85.61 %, and an F_1 score of 84.89 %. Fig. 3 presents a visual assessment of the model performance, featuring three randomly selected images from the test dataset. The F_1 score was used to evaluate the balance between precision and recall, offering a single metric that reflects the overall accuracy of the model in identifying segmented regions.

Fig. 4 provides a detailed comparison between the ground truth and various U-Net segmentation methods for a nanoparticle region within the original image.

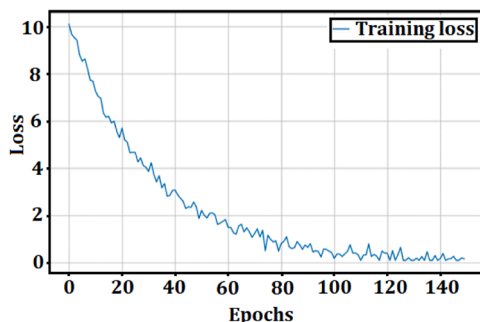


Fig. 2. Training loss curves showing the progression of U-Net training.

3.2. Quantitative analysis of nanoparticle area distribution

The segmented images generated by the U-Net model were further analyzed using image processing techniques to accurately identify and measure the nanoparticle regions. Contours representing each white region in the segmented images were detected, and the areas of these regions were calculated in pixels. These pixel-based measurements were then converted to actual nanoparticle areas using the image resolution of 135 pixels per micrometer. This analysis enabled the determination of the minimum, maximum, and average diameters of nanoparticles for each category, based on the percentage of added NZ: (a) 2.5 wt%, (b) 5 wt%, and (c) 7.5 wt%. The summary results of this analysis are presented in Table 2.

3.3. Characterization of nanocomposites

To investigate the distribution and presence of nano-zeolite within the polymer matrix, SEM images of the fractured surfaces were captured. Fig. 5 shows the fractured surfaces of PA6/NZ nanocomposites with different NZ contents (2.5, 5, and 7.5 wt%). A key observation from Fig. 5 is the uniform dispersion of nanoparticles across all compositions, regardless of the NZ content. As the concentration of NZ increases, the density of nanoparticle distribution significantly rises. This effective dispersion can be attributed to the strong affinity between the polymer and nanoparticles, which, during mixing, breaks down particle clusters and facilitates the homogeneous dispersion of nanoparticles within the matrix.

However, at higher NZ concentrations, some clusters (indicated by yellow arrows in Figs. 5b and 5c) and aggregates (highlighted by red arrows) become visible. These agglomerates result from extensive particle clustering, leading to aggregation on a micrometer scale. The increase in particle count, under constant processing time and mixing energy, reduces the available external energy per particle to overcome interparticle van der Waals forces, which hinders uniform distribution and results in random clusters in the final sample. On the other hand, a higher particle count increases the frequency of inter-particle collisions during processing, which promotes particle dispersion. This phenomenon is evident in Figs. 5b and 5c, where a significant number of finely dispersed nanofillers are observed throughout the polymer matrix. The final distribution of nanoparticles within the matrix is ultimately determined by a delicate balance between these opposing factors, alongside the polymer-filler affinity, which favors dispersion.

3.4. Hardness evaluation

Fig. 6 shows the hardness values obtained for PA6 and PA6 nanocomposites. The findings indicate a notable enhancement in the hardness of the nanocomposites when nanoparticles are incorporated into the PA6 matrix. Specifically, the hardness increases by over 7 % with the addition of 7.5 % NZ. This improvement is primarily due to the formation of microstructural bonds between the base matrix and the added nanoparticles on the sample surface [25]. These bonds effectively resist applied surface forces, demanding greater energy for breakage, which contributes to the observed increase in hardness. Furthermore, the presence of nanoparticles limits the mobility of polymer chains, leading to an intrinsic rise in both material stiffness and hardness [26].

The steady improvement in hardness with the addition of nanofillers highlights the incremental reinforcing effect of nanoparticles with filler additions [27]. This increase in hardness is anticipated to enhance the wear resistance of the nanocomposite samples. However, when increasing the nanofiller content from 5 to 7.5 wt%, the rate of hardness improvement slows compared to lower concentrations. This may be attributable to increased particle clustering and aggregation within the matrix, which can disrupt dispersion. Poor dispersion reduces the nanoparticles effectiveness in enhancing rigidity and forming strong polymer-filler bonds that resist the forces applied during hardness

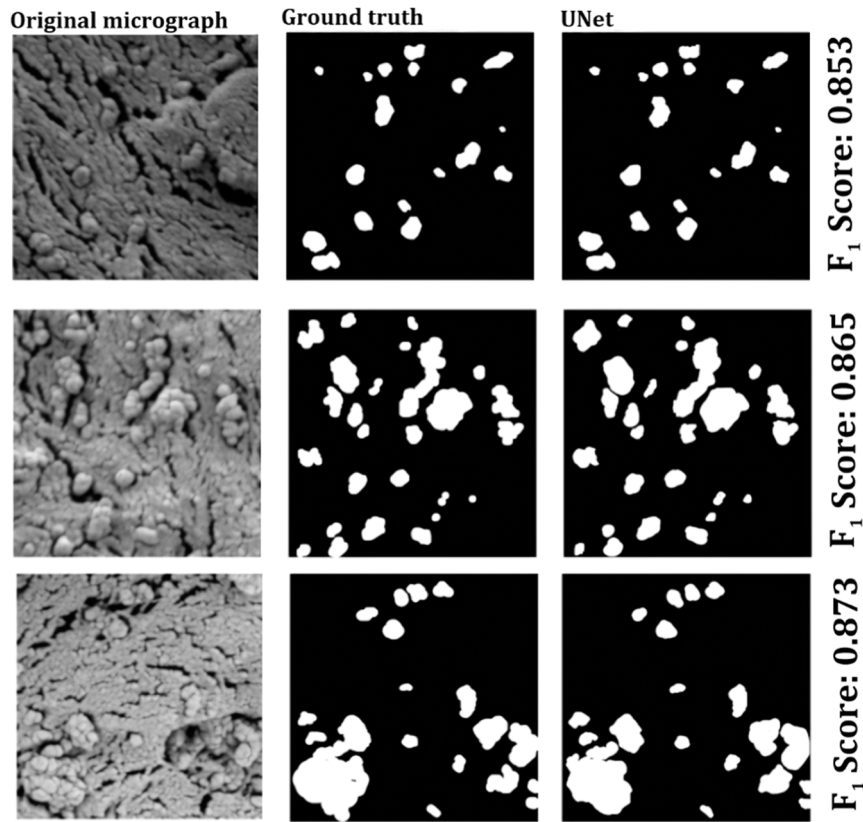


Fig. 3. Visual evaluation of the trained U-Net model, illustrating three randomly selected images from the test dataset.

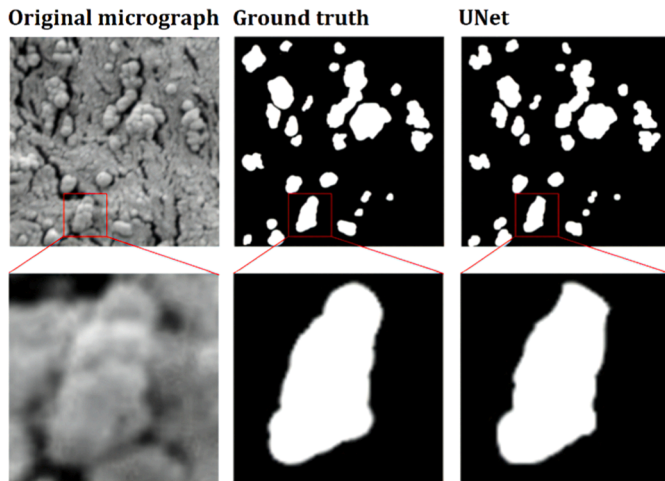


Fig. 4. Example of nanoparticle segmentation in the nanocomposite.

Table 2

Quantified outcomes of nanoparticle size distribution analysis.

Sample code	Zeolite (wt%)	D_{min} (μm)	D_{max} (μm)	D_{ave} (μm)
PA6/2.5NZ	2.5	0.0628	0.2456	0.09658
PA6/5NZ	5	0.0645	0.3060	0.1395
PA6/7.5NZ	7.5	0.0786	0.6907	0.2141

D_{min} : minimum diameter, D_{max} : maximum diameter, D_{ave} : average diameter

testing. Therefore, achieving proper nanoparticle dispersion within the polymer matrix is crucial for improving mechanical properties. Uniform dispersion facilitates optimal load distribution, thereby enhancing hardness.

3.5. Wear characteristics

Fig. 7 illustrates the changes in the friction coefficient for pure PA6 and its nanocomposites under applied loads of 60 N and 100 N. It is observed that the friction coefficient increases for all tested materials as the applied load increases. This rise is attributed to changes in the actual contact area under the normal load. The surface skin formed during injection molding significantly influences the friction behavior of both PA6 and its nanocomposites. This surface skin, which is harder than the inner core, differs in composition from the bulk material [11]. As a result, it affects the break-in behavior, where higher loads shorten the sliding distances required for break-in.

As shown in Fig. 7, the addition of NZ leads to a reduction in the friction coefficient, with the lowest value recorded for PA6/5NZ. The semi-spherical geometry of NZ minimizes direct polymer contact in the wear zone, functioning as a solid lubricant between the contacting surfaces. This reduction in contact area contributes to the observed decrease in the friction coefficient. Similar results were reported by Mohsenzadeh et al. [28], who studied the wear behavior of PA6-PP nanocomposites incorporating CaCO_3 nanoparticles. In the polymer matrix, NZ not only acts as a self-lubricant but also exhibits nucleating effects, potentially increasing the degree of crystallinity. An increase in crystallinity typically leads to a harder surface and increased material stiffness, which in turn results in higher surface resistance [29]. Consequently, polymers with higher crystallinity demonstrate greater resistance to frictional forces. Additionally, inorganic nanoparticles like NZ, which have lower friction coefficients compared to organic polymers, can effectively reduce the overall friction coefficient in

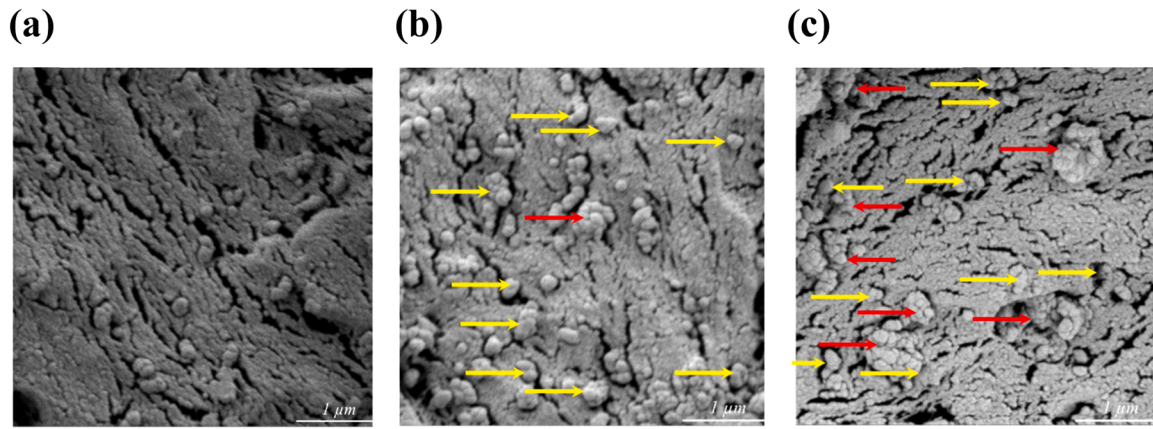


Fig. 5. SEM images of fractured surfaces in nanocomposites with different nano-zeolite weight percentages: (a) 2.5 wt%, (b) 5 wt%, and (c) 7.5 wt%.

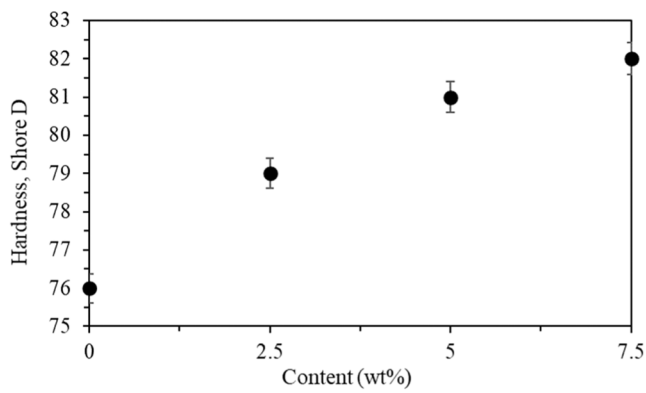


Fig. 6. Shore hardness values of PA6/NZ nanocomposites with varying nano-zeolite contents.

nanocomposite systems [30].

Fig. 8 illustrates the temperature variations observed during the wear test under both 60 N and 100 N loads. Initially, all specimens, including pure PA6 and the nanocomposites, experienced a rapid temperature increase under both loads. The observed increase in temperature can be attributed to initial frictional heating at the surface of the samples. In this phase, the heat generated by friction surpasses the rate of heat transfer via conduction from both the contact interface and the bulk material, as well as through convection. As the experiment

progresses and the system nears thermal equilibrium, the temperature stabilizes for all samples except for neat PA6 under a load of 60 N, which demonstrated a gradual rise in surface temperature throughout the test. This stabilization suggests that the mechanisms of heat generation and transfer achieved a balance, allowing most samples to reach a steady-state thermal condition. Under the 100 N load, all samples showed a continuous rise in surface temperature due to the increased normal force and the resultant friction-induced heating. However, the temperature increase in PA6 was more pronounced and somewhat unstable, particularly towards the end of the test, as depicted in Fig. 8b. This rapid temperature rise, attributed to the low thermal resistance and conductivity, can accelerate surface wear and increase frictional forces in pure PA6.

Fig. 8 illustrates that the incorporation of NZ into PA6 leads to a reduction in temperature compared to the unreinforced sample under both loads. This temperature decrease with nanoparticle addition is likely due to the superior thermal conductivity of the inorganic nanoparticles relative to pure PA6. The high specific surface area of these nanofillers [27] allows for significant surface interaction with the polymer matrix, enhancing heat transfer and reducing temperature build-up on the polymer sample surface [31,32]. However, as the nanoparticle concentration increases from 5 to 7.5 wt%, the thermal performance of the nanocomposite declines, resulting in a higher surface temperature during testing. This diminished performance can be attributed to the less uniform dispersion of particles and the formation of localized clusters in the PA6/7.5NZ sample. This poor spatial distribution of NZ nanoparticles can result in a non-uniform temperature

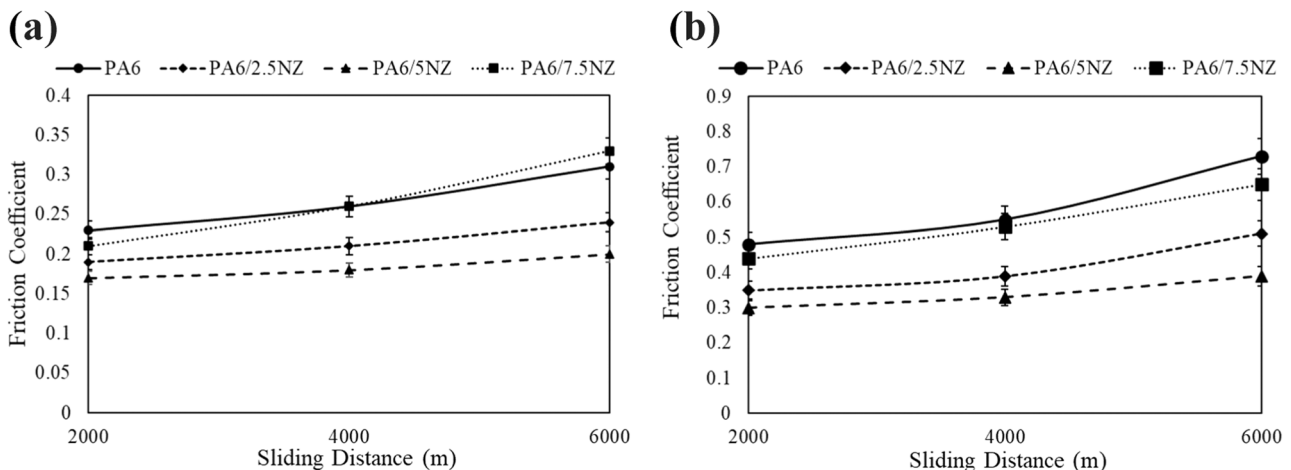


Fig. 7. Variation of the friction coefficient for the compounds under: (a) 60 N load, and (b) 100 N load.

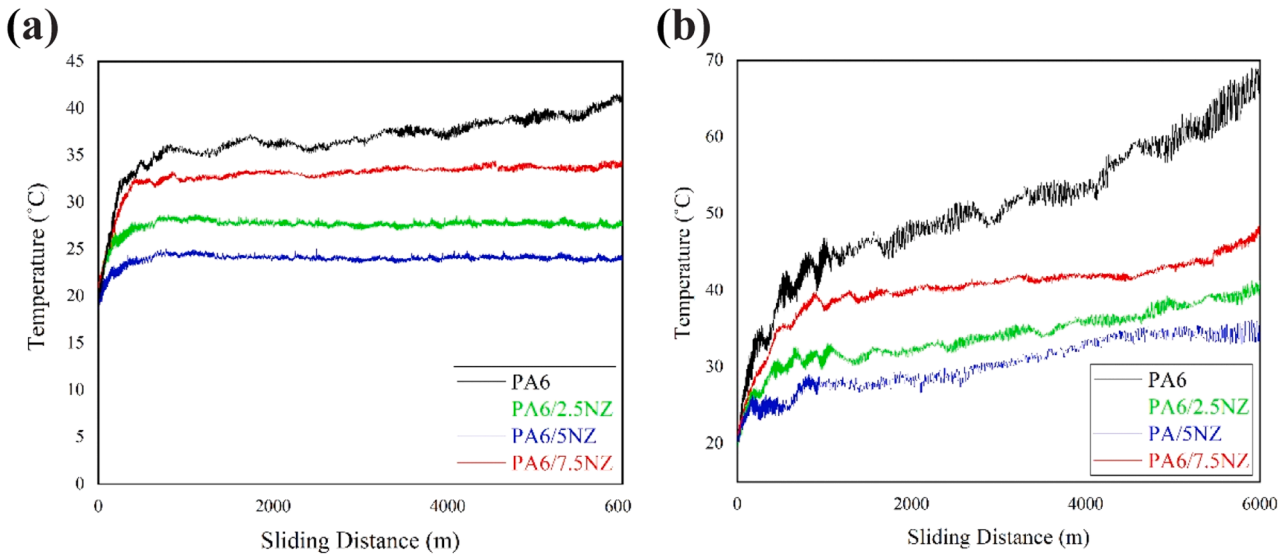


Fig. 8. Temperature variation with sliding distance for different samples under: (a) 60 N load, and (b) 100 N load.

distribution under normal load and contribute to local heat accumulation on different regions of the sample surface. Although the temperature increases by 7.5 wt% compared to lower levels of NZ inclusion, this rise remains more controllable and stable than in pristine PA6, where the surface temperature exhibits an exponential increase during testing.

The impact of applied load on the specific wear rate is illustrated in Fig. 9, which shows the variation in specific wear rate for different compositions after testing. As depicted, the applied load has a direct influence on the wear rate across all compositions [33]. For instance, increasing the load from 60 N to 100 N results in a 120 % rise in the specific wear rate for neat PA6. It is clear from Fig. 9 that PA6 nanocomposites exhibit a lower wear rate than neat PA6 at all investigated normal loads. This enhanced wear resistance can be attributed to the increased surface resistance against normal loads, where the addition of NZ improved surface integrity, hardness, and stiffness. As shown in Fig. 9, incorporating 5 wt% NZ, which is identified as the optimal concentration, leads to a 43 % reduction in wear rate at 60 N and a 57 % reduction at 100 N, compared to pure PA6. This significant reduction indicates a robust interaction between the reinforcing nanoparticles and the polymer matrix, as well as optimal dispersion within the matrix, providing a substantial specific surface area between these nanoparticles and the polymer matrix [34].

NZ nanoparticles also function as a lubricant, owing to their semi-spherical geometric shape. These nanoparticles act as a third body, positioning themselves between the sample surface and the loading

device, thus preventing direct contact and sliding between wear surfaces [17,35,36]. Moreover, the incorporation of NZ into the polymer matrix may induce nucleation effects, potentially enhancing the degree of crystallinity and surface hardness [37]. This is evident in Fig. 6, where the inclusion of NZ increases surface hardness across all nanocomposite samples. The reduction in the friction coefficient, shown in Fig. 7, along with the decreased temperature at the contact surface between the pin and disk, as depicted in Fig. 8, both contribute to the observed wear reduction. However, as indicated in Fig. 9, increasing the NZ content beyond optimal levels (i.e., above 5 wt%) results in diminished wear resistance compared to lower concentrations. This increase in wear rate is attributed to particle clustering within the polymer matrix, leading to less uniform particle dispersion [38–40]. These clusters can act as abrasive agents, exacerbating wear [41].

3.6. Numerical assessment

To assess the impact of NZ dispersion, load intensity, and surface hardness on the friction coefficient and specific wear rate, a regression modeling was developed using support vector regression (SVR). SVR, an advanced technique derived from support vector machines (SVM), is effective for regression tasks by predicting continuous variables. It excels in capturing complex and nonlinear relationships due to its use of kernel functions, which allow it to model intricate interactions and provide reliable predictions even with limited datasets, as often

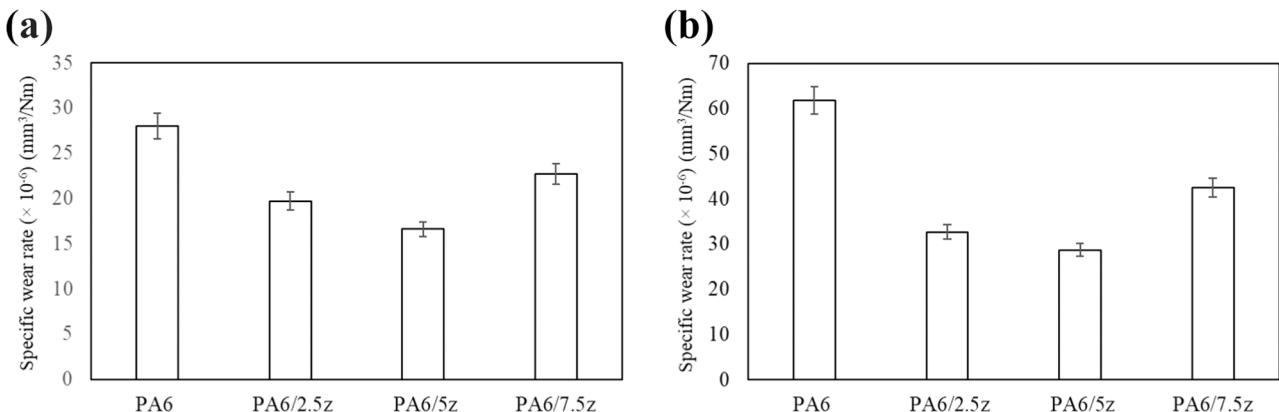


Fig. 9. Specific wear rate of samples during wear tests under: (a) 60 N load, and (b) 100 N load.

encountered in experimental research. To build the model, it is first necessary to establish a metric for quantifying particle dispersion based on the particle diameters outlined in Table 1.

3.6.1. Dispersion quantification

The nanocomposite formulation, as illustrated in Figs. 7 and 9, significantly influences both the friction coefficient and the specific wear rate. This observation underscores the critical role that dispersion characteristics play at various loadings in determining these outcomes. Effective dispersion in nanocomposites is essential for fully leveraging the properties of nanoparticles throughout the matrix, thereby enhancing the material overall performance. In contrast, poor dispersion can lead to nanoparticle agglomeration, which diminishes the reinforcement effectiveness and may introduce defects or weaknesses in the material. Therefore, incorporating a quantified measure as an input variable in the model is vital for assessing dispersion quality and understanding its impact on key tribological properties, such as the friction coefficient and specific wear rate.

In this study, a Combined Dispersion Index (CDI) was developed to evaluate the uniformity of NZ nanoparticle dispersion within the polymer matrix. The CDI is derived from the average, minimum, and maximum diameters of the dispersed nanoparticles, as presented in Table 2. By incorporating these three parameters, CDI serves as a comprehensive metric that reflects the state of dispersion in terms of both homogeneity and uniformity. CDI integrates two distinct indices: the Range-based Dispersion Index (RDI) and the Relative Span Index (RSI). The RDI offers a dimensionless measure of particle size distribution by normalizing the range relative to the average particle size, while the RSI accounts for the normalized spectrum from $D_{\min}$ to $D_{\max}$, representing dispersion based on a relative span. Consequently, CDI can be expressed as follows:

$$CDI = \underbrace{\frac{D_{\max} - D_{\min}}{D_{ave}}}_{\text{Range-based Dispersion Index (RDI)}} \times \underbrace{\frac{D_{\max} - D_{\min}}{D_{\max} + D_{\min}}}_{\text{RelativeSpanIndex (RSI)}} \quad (6)$$

The incorporation of the CDI in the model facilitates establishing a correlation between dispersion characteristics and the observed tribological behavior. Lower CDI values suggest a narrower particle size distribution, indicating higher size uniformity, whereas higher CDI values imply reduced uniformity due to a broader range of particle sizes.

3.6.2. Dataset preparation

Following the quantification of dispersion using the CDI metric, it is now feasible to develop a model utilizing SVR. In this model, the CDI, load, and surface hardness will serve as input variables, while the tribological properties, including the friction coefficient and specific wear rate, will act as response variables. The hardness, friction coefficient (measured at a sliding distance of 6000 m), and specific wear rate were obtained from the data presented in Figs. 6, 7, and 9, respectively. Table 3 presents the dataset used in this study, which comprises a combination of experimental and numerical outcomes.

Table 3
Dataset used for constructing the regression model.

Test No.	Input variables			Tribological response	
	CDI	Load (N)	Hardness, Shore D	Friction coefficient	Specific wear rate (mm ³ /Nm)
1	1.121891	60	79	0.24	1.97386×10^{-5}
2	1.128423	60	81	0.2	1.65692×10^{-5}
3	2.274743	60	82	0.33	2.26608×10^{-5}
4	1.121891	100	79	0.51	3.27485×10^{-5}
5	1.128423	100	81	0.39	2.86550×10^{-5}
6	2.274743	100	82	0.65	4.25439×10^{-5}

3.6.3. Model validation

Fig. 10 demonstrates the performance of the SVR model in predicting the friction coefficient and specific wear rate using experimental data. To evaluate the regression model accuracy, two metrics were employed: mean squared error (MSE) and the coefficient of determination (R^2). The $Y = X$ line acts as a reference for model accuracy; ideally, the red points (predicted responses) should lie directly on this dashed line, signifying a perfect match between model predictions and experimental results. Any deviations from this line are captured in the performance metrics.

According to Fig. 10a, the R^2 value of 0.9592 indicates that the SVR model covers 95.92 % of the variance in the friction coefficient, showing strong predictive power. The low MSE of 0.0010 further confirms the model accuracy, with minimal discrepancies between predicted and actual values. For the specific wear rate in Fig. 10a, an R^2 of 0.8720 suggests the model includes 87.20 % of its variance, indicating a strong relationship with the input variables, though with slightly more unexplained variability. The near-zero MSE for the specific wear rate highlights the model excellent predictive performance.

3.6.4. Response surface analysis

The construction of the model enables the generation of 3D surface response plots, depicting how each input variable influences the resulting tribological performance within a predefined range. In practical scenarios, generating these plots requires a large dataset. Fig. 11 presents these three-dimensional plots, showing how two tribological characteristics vary with respect to the input variables plotted on the X and Y axes. For each plot, the third variable was held constant at its average value across different compounds. Upon initial examination, it is evident that the dependency curves for both outputs are identical in shape, indicating a strong correlation between the friction coefficient and the specific wear rate across the different samples.

Figs. 11a and 11b demonstrate that higher CDI values, indicative of inferior particle size uniformity, are associated with increased friction coefficients and specific wear rates. This suggests that suboptimal dispersion uniformity significantly influences the tribological behavior of the sample. The data reveal that the relationship between CDI and the tribological properties shows a steeper gradient at lower loads, such as 60 N, and a more gradual slope at higher loads, such as 100 N. At lower loads, the impact of CDI on tribological performance is more pronounced; higher CDI values, reflecting worse dispersion, lead to greater friction and wear due to the potential presence of abrasive nanoparticle clusters. Conversely, as the load increases, the effect of CDI diminishes. Higher loads intensify stress and introduce additional factors, such as surface deformation and material bulk properties, which overshadow the influence of dispersion. Thus, while poor dispersion continues to affect friction and wear, the material response becomes less sensitive to CDI at higher loads.

Figs. 11c and 11d depict the variation in tribological properties with respect to hardness and load at an average CDI of 1.5084. Unlike CDI and load, hardness has a minimal effect on friction and wear due to its limited role in resisting mechanical and thermal loads. As shown in Figs. 11c and 11d, the friction and wear follow a parabolic trend with increasing hardness at a load of 60 N, reaching a minimum around a hardness value of 80. This behavior arises from a balance between surface deformation and adhesion: lower hardness increases adhesion and friction due to greater surface softness, whereas higher hardness reduces deformation but introduces brittleness, which can elevate both wear and friction. The parabolic trend reflects the interaction between these factors, with minimal friction occurring at an optimal intermediate hardness level. At higher loads, such as 100 N, the parabolic relationship between friction and hardness becomes less pronounced compared to lower load intensities. This can be attributed to the intensification of wear and contact mechanisms, which diminish the influence of hardness on the overall tribological performance. Increased contact pressure at higher loads leads to more significant surface deformation. At lower hardness values, this results in heightened plastic deformation,

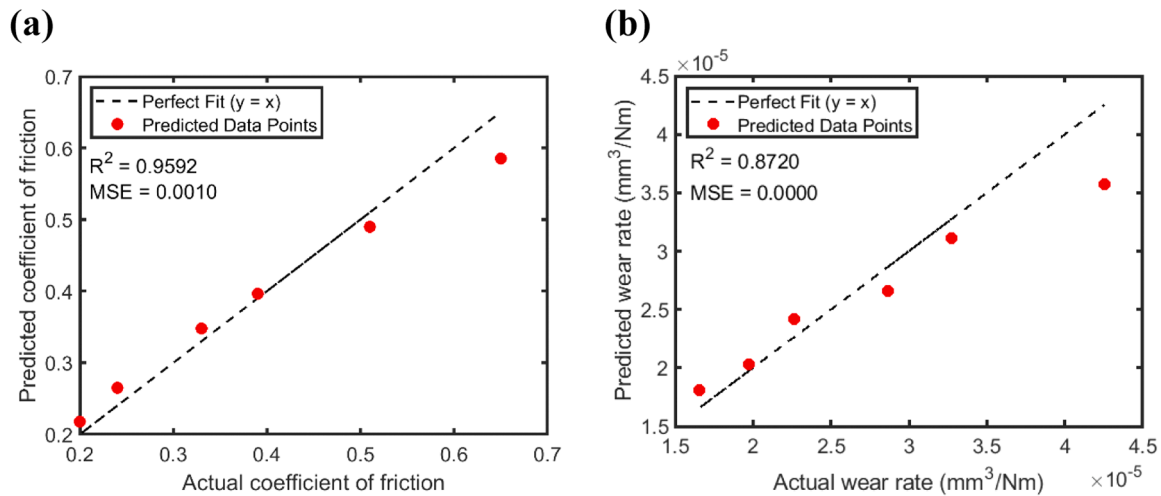


Fig. 10. Correlation plots between predicted and actual values for: (a) friction coefficient, and (b) wear rate, using the SVM model.

increasing the friction coefficient. As hardness increases, the material better resists deformation, but interactions between surface asperities and the potential for micro-fractures still contribute to elevated friction. As a result, hardness variations have a reduced impact on friction compared to lighter loading conditions, where surface hardness plays a more critical role in determining tribological behavior.

Figs. 11c and 11d show how tribological properties vary with hardness and CDI at an average load of 80 N. At low CDI values, the friction coefficient and wear rate exhibit a parabolic trend, reaching a minimum at a hardness of around 80. This optimal hardness balances particle size uniformity and material properties, minimizing friction and wear. Deviations from this optimal hardness result in increased friction and wear, either due to material deformation or reduced compliance, which intensifies abrasive wear and interactions between surface asperities. At higher CDI values, indicating poorer dispersion quality, the relationship becomes less symmetric and shifts to a rising trend. This is because poor dispersion causes larger abrasive clusters, leading to a continuous increase in friction and wear with hardness. This shift highlights the greater impact of dispersion quality on tribological performance at higher hardness levels.

3.6.5. Sensitivity analysis

To perform a sensitivity analysis of the input parameters on tribological responses, the effects of each parameter on the friction coefficient and specific wear rate were examined individually while maintaining the other two input variables at their mean values. Fig. 12 illustrates the relevant plots, suggesting that variations in load have the most significant impact on both responses, with CDI being the second most influential parameter and surface hardness having the least effect. Additionally, except for hardness, which causes a parabolic variation in friction and wear, the other two parameters exhibit a continuously increasing trend. This indicates that dispersion quality and load intensity play a crucial role in modifying the friction coefficient and specific wear rate.

3.7. Microstructural analysis of worn surfaces

SEM micrographs of the worn surfaces were analyzed to investigate the predominant wear mechanisms across various compounds. The wear morphology was examined, and the results are presented for a lower load of 60 N in Fig. 13 and a higher load of 100 N in Fig. 14.

Fig. 13 illustrates the worn surfaces subjected to a 60 N load. Fig. 13a reveals abrasive wear features, such as grooves aligned with the sliding direction, indicated by red arrows. On the surface of pure PA6 (Fig. 13a), severe wear is characterized by deep grooves resulting from the disc

action, attributed to the softness of the unreinforced polymer surface. The increased surface temperature due to friction leads to the breakdown of polymer chain bonds, which exacerbates wear. Furthermore, material removal during surface wear can lead to debris accumulation and the formation of larger chips, intensifying wear.

Fig. 13b-d illustrate that the incorporation of NZ into pure PA6 results in a smoother worn surface compared to Fig. 13a. The worn surfaces of nanocomposites containing 2.5 and 5 wt% NZ show fewer signs of severe abrasive wear. This indicates that the presence of nanoparticles promotes a transition from sliding to rolling contact. This transition is attributed to enhanced molecular chain mobility of the polymer matrix at high temperatures, which can reduce the polymer-nanoparticle interfacial adhesion. As a result, nanoparticles can move more freely, facilitating the shift from sliding to rolling contact. This rolling effect diminishes the contact area, a phenomenon supported by previous studies in various polymer applications [42–44]. Additionally, the decrease in friction coefficient, temperature, and wear, as shown in Figs. 7–9, further corroborates this effect. Conversely, Fig. 13d reveals that adding 7.5 wt% NZ leads to a more pronounced wavy surface, with higher abrasive wear tracks and detached debris, resulting in small surface pits. This observation is consistent with SEM images in Fig. 5c, which show the presence of large clusters. These clusters contribute to a non-uniform dispersion at the contact surface between the pin and disc, leading to exacerbated wear due to the formation of three-body wear conditions.

Fig. 14 shows the worn surfaces subjected to a 100 N load. Under this increased load, all surfaces exhibit a distinctly striated pattern, characterized by wider grooves and more pronounced wear lines compared to surfaces tested at 60 N. At lower loads, as seen in Fig. 13, minor abrasion was the dominant wear mechanism. In contrast, severe abrasion and scuffing become the primary wear mechanisms at higher loads. Due to its lower resistance to plastic deformation, PA6 displays deeper abrasion marks, as evidenced by the extensive wear lines in Fig. 14a. Fig. 14a also highlights scuffing, marked by deep and wide grooves predominantly aligned with the sliding direction.

Zhang et al. [45] observed that frictional heat generated during sliding can significantly increase the contact temperature, leading to a reduction in mechanical properties such as hardness and shear resistance. This occurs due to intensified plastic flow at the interface. The formation of grooves is likely the result of cutting by hard asperities from the counterface and the presence of third-body polymeric debris in the contact zone. During reciprocating sliding, cyclic plastic deformation can cause work hardening, increasing the hardness of polymeric debris. This hardened debris may act as an abrasive agent in subsequent sliding cycles, exacerbating wear. Additionally, the presence of adhesive

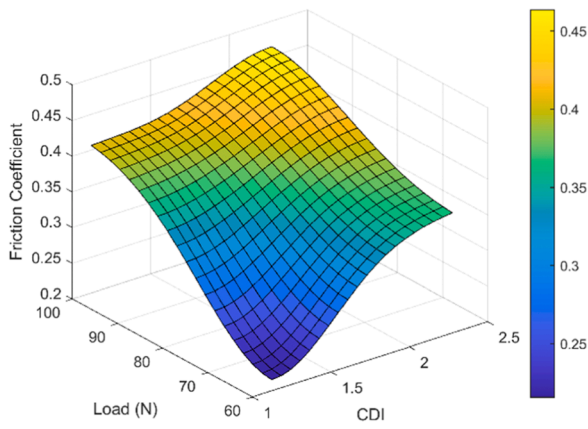
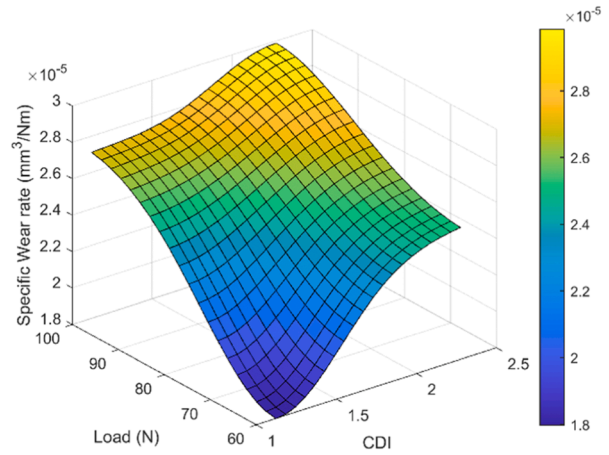
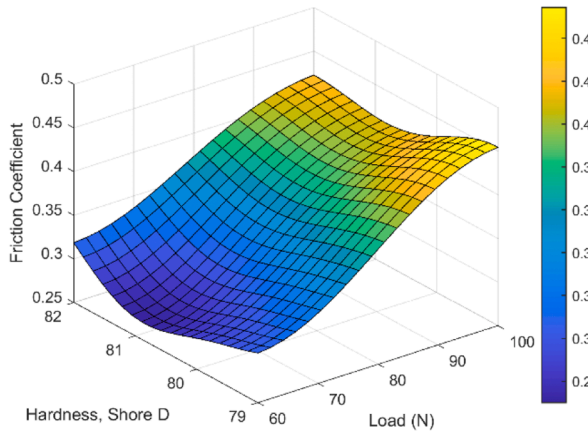
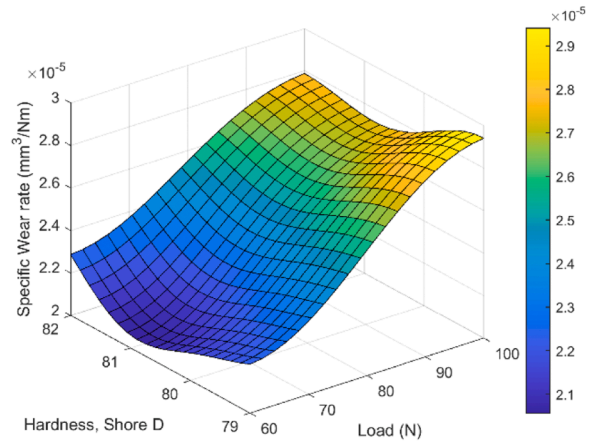
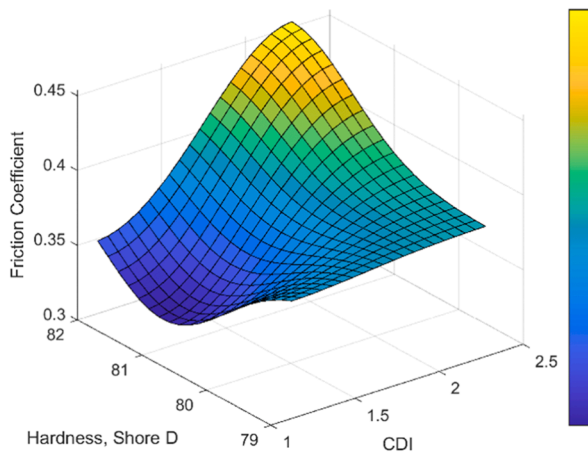
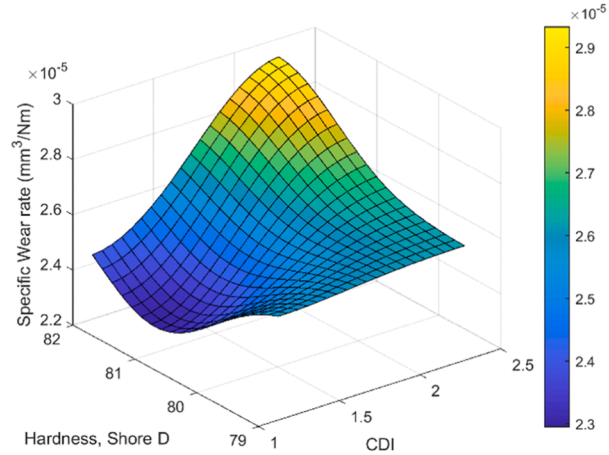
(a) Hardness = 80.67**(b) Hardness = 80.67****(c) CDI = 1.5084****(d) CDI = 1.5084****(e) Load = 80 N****(f) Load = 80 N**

Fig. 11. 3D plots illustrating the variation of the friction coefficient and specific wear rate with respect to CDI, load, and hardness (the third variable is set to the average value across all nanocomposites).

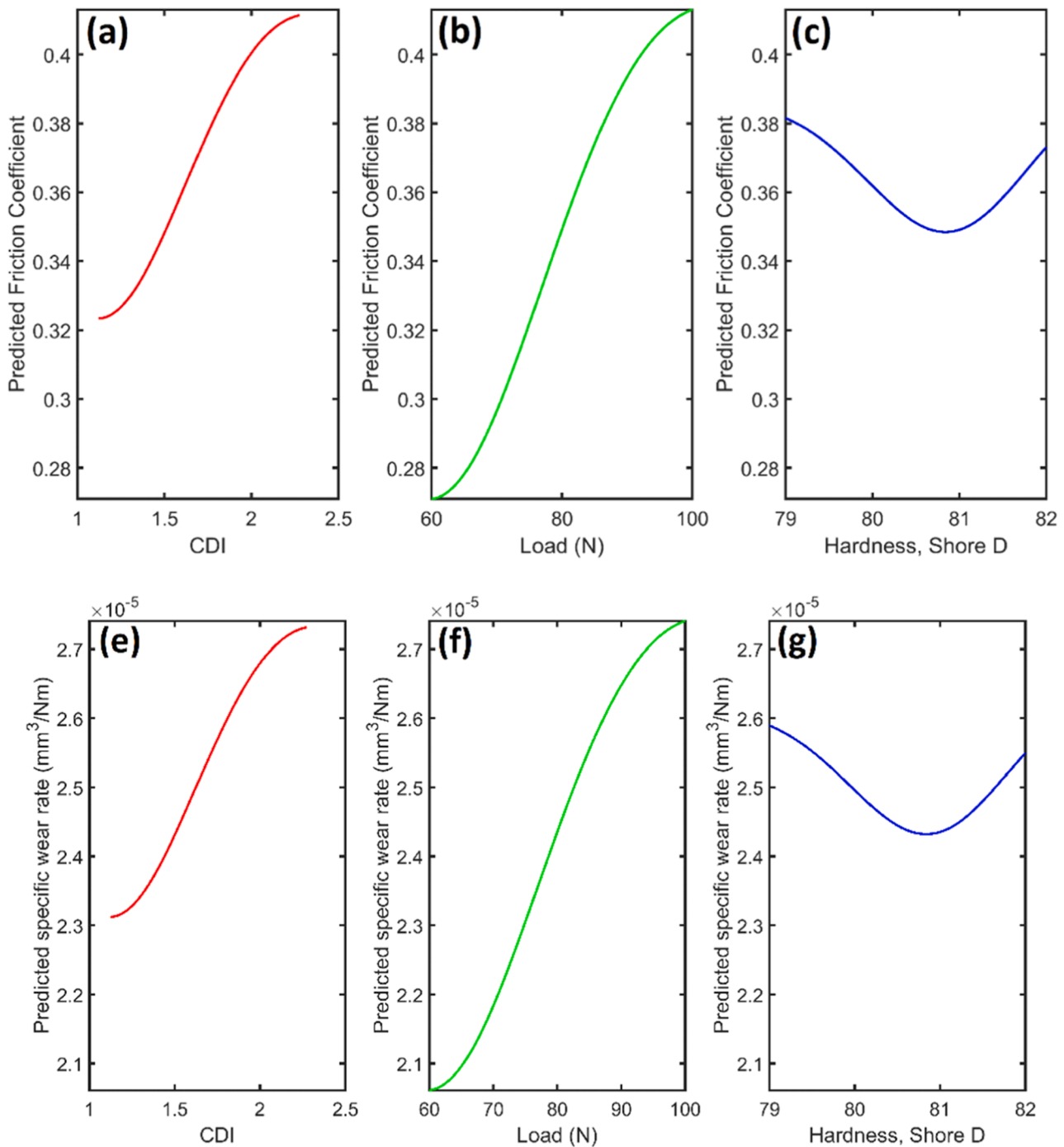


Fig. 12. Sensitivity analysis of friction coefficient and specific wear rate as output variables, in response to CDI, load, and hardness as input factors.

wear on the worn surface, as seen in Fig. 14a, is likely a result of material softening, segmentation, and the adhesion of dislodged material in the worn zone, driven by the high temperatures and contact stresses [46].

As shown in Fig. 14b, the worn surface of the PA6/2.5NZ sample exhibits significantly narrower grooves compared to pure PA6 in Fig. 14a, with minimal evidence of scuffing. However, traces of abrasive wear and sliding marks remain visible. The reduced wear damage on the PA6/2.5NZ surface, relative to pure PA6, can be attributed to the enhanced wear resistance provided by the addition of NZ. Severe ploughing is largely suppressed in this material, although some surface damage is still present, as shown in Fig. 14b. The worn surface of the PA6/5NZ sample demonstrates an even smaller wear track than the other specimens, with no visible scuffing and a much smoother surface,

as seen in Fig. 14c. Despite its superior endurance compared to PA6 and PA6/2.5NZ, minor abrasion tracks with shallow grooves are still observed on the PA6/5NZ surface, indicating that the inclusion of NZ effectively mitigates thermal effects. This improvement is likely due to the thermal barrier properties of NZ nanoparticles, which reduce frictional heat generation and help dissipate heat, particularly at elevated temperatures, as shown in Fig. 8b.

In the PA6/7.5NZ sample shown in Fig. 14d, micro-cracks, scuffing, and ploughing grooves are more visible compared to samples with lower NZ content. However, these wear features are more localized and less widespread than those observed in neat PA6, as depicted in Fig. 14a. The presence of flaky debris and micro-cracks in Fig. 14d indicates a shift in wear mechanisms, likely towards micro-cracking or surface fatigue, with

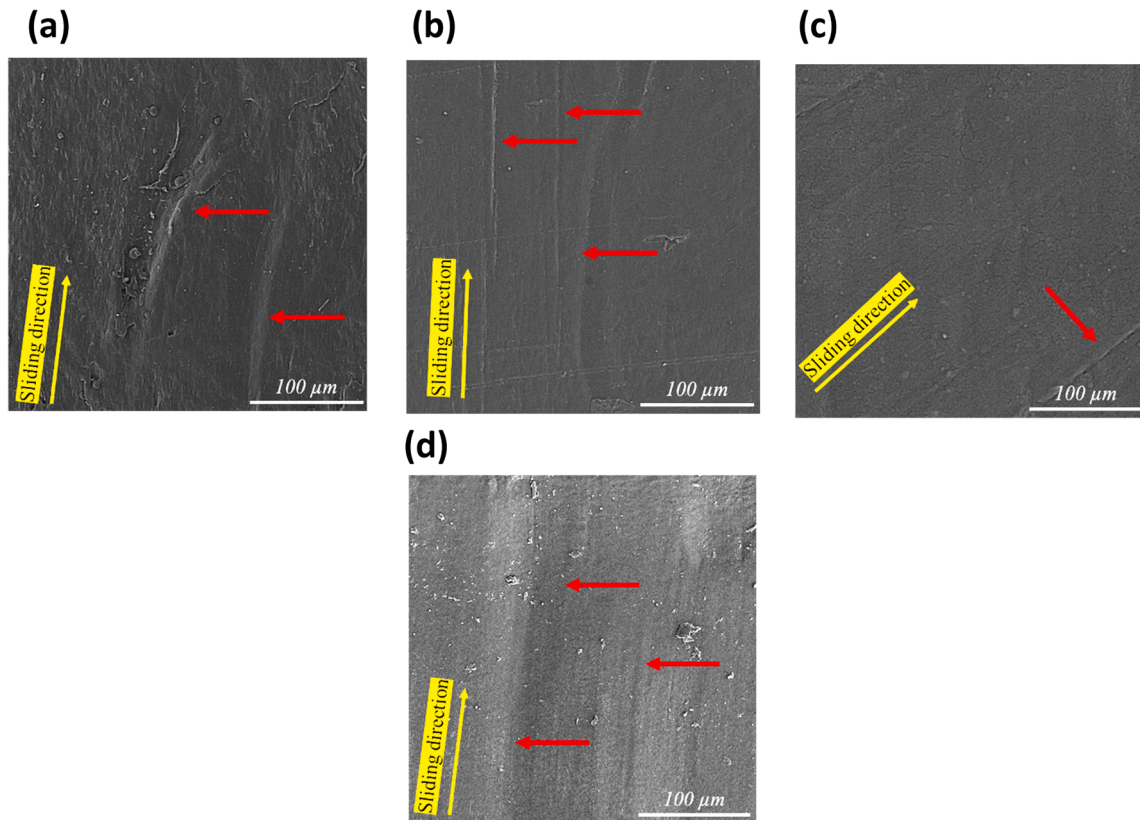


Fig. 13. SEM images of worn surfaces from samples tested under a 60 N load: (a) Neat PA6, (b) PA6/2.5NZ, (c) PA6/5NZ, and (d) PA6/7.5NZ.

reduced evidence of thermally induced damage such as surface yielding and adhesive wear, which were more prevalent in pure PA6, as seen in Fig. 14a.

4. Limitations, challenges, and future scope

This study aimed to establish a correlation between the microstructural features of PA6/NZ nanocomposites and their tribological properties through the integration of computer vision, machine learning, and experimental testing. While the results provide valuable knowledge, certain limitations need to be addressed, along with directions for future research.

Although the U-Net-based segmentation method effectively quantified nanoparticle distribution, image complexity and noise may have introduced uncertainties in the dispersion measurements. Future research could employ more advanced image processing techniques or deep learning models with improved robustness against artifacts and noise to enhance segmentation accuracy and dispersion quantification.

The study was confined to specific NZ concentrations (2.5, 5, and 7.5 wt%) and load conditions (60 N and 100 N), potentially limiting its ability to cover the full range of tribological behaviors under different environmental and operational conditions, such as temperature and humidity. Expanding the range of NZ concentrations and examining performance under diverse environmental conditions would provide a deeper understanding of the material behavior across broader applications.

Furthermore, only pin-on-disk wear tests were conducted, which may not fully include all wear mechanisms relevant to practical applications. Future studies may consider alternative wear testing methods, such as reciprocating sliding or abrasive wear tests, to better simulate real-world conditions and broaden the assessment of wear mechanisms.

The study's focus on NZ as the sole reinforcement material may have constrained the exploration of other nanofillers or hybrid composites

that could be characterized by superior tribological properties. Investigating additional nanofillers or hybrid composites could further optimize the mechanical and tribological performance of the nanocomposite, expanding its industrial applicability.

While the SVR model demonstrated superior predictive performance, it may not fully account for the intricate relationships between factors influencing tribological performance. Future research might benefit from employing more sophisticated machine learning models, like neural networks, to enhance predictive accuracy and gain a better grasp of the connections between nanoparticle distribution, load conditions, and material properties.

This study primarily examined the morphology of nanocomposites; however, the microscopic interactions at the polymer-steel interface could significantly affect wear behavior. Therefore, it is essential to analyze the adhesion properties and wear morphology between the polymer nanocomposite and the steel disk surface. Future research may focus on investigating these adhesion effects and the resulting wear morphology on the steel surface to elaborate on the tribological performance of polymer-steel interfaces.

5. Conclusion

This study aimed to correlate the wear properties of PA6/NZ nanocomposites with their morphological characteristics through a combination of visual, experimental, and numerical evaluation techniques. Pin-on-disk tests were conducted to measure surface temperature, friction coefficient, and specific wear rate under applied loads of 60 N and 100 N. A U-Net-based automated image processing technique was developed and applied to SEM micrographs to analyze the microstructure by detecting and segmenting nanoparticles to compute the minimum, average, and maximum particle diameters. The CDI metric was then used to quantify nanoparticle dispersion, which was subsequently correlated, along with loading intensity and surface hardness, to

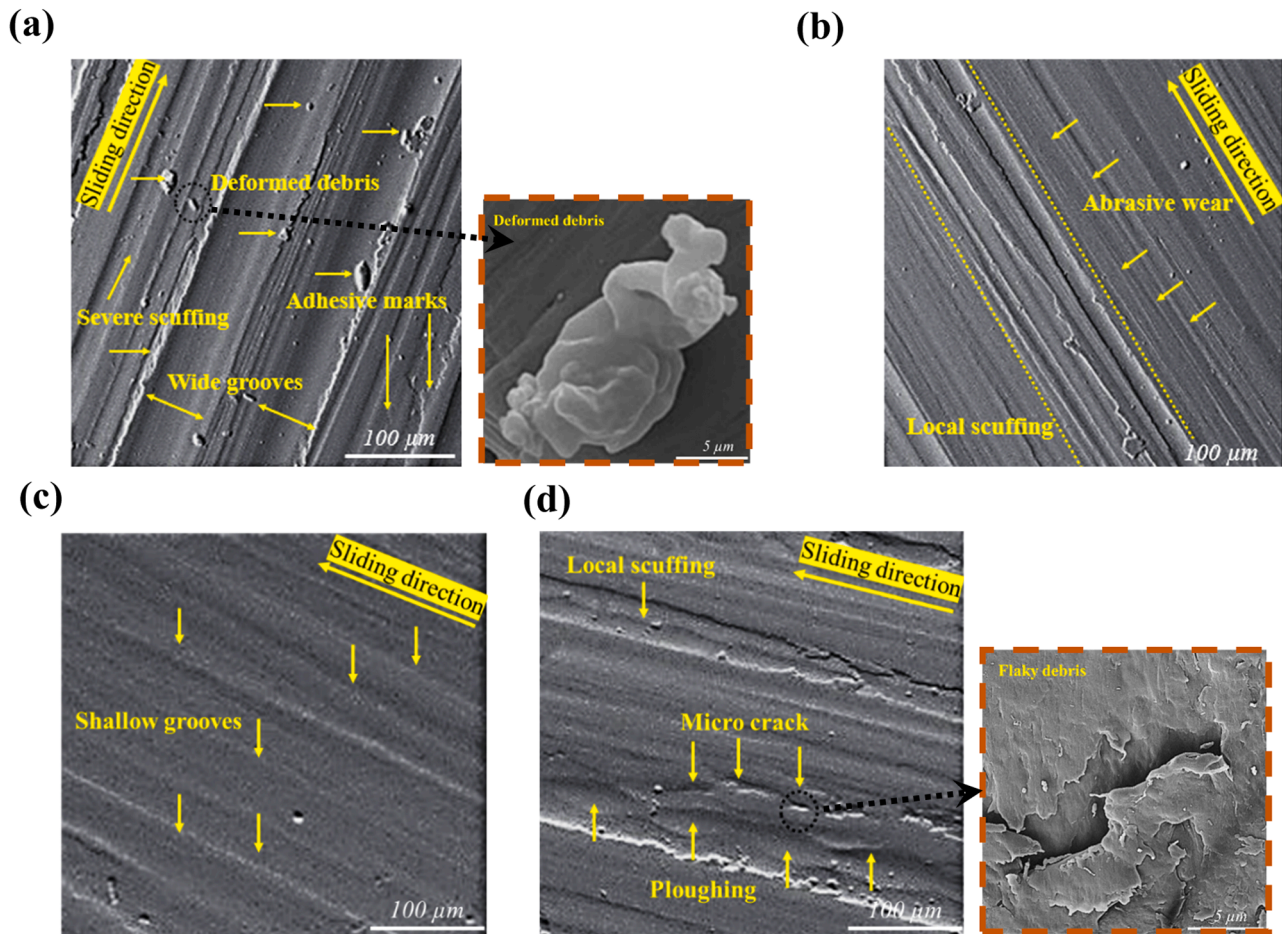


Fig. 14. SEM images of worn surfaces from samples tested under a 100 N load: (a) Neat PA6, (b) PA6/2.5NZ, (c) PA6/5NZ, and (d) PA6/7.5NZ.

tribological performance using a SVR model. The microstructural features of the worn surfaces were further examined via SEM to identify the dominant wear mechanisms in the nanocomposite samples.

Qualitative analysis of the SEM micrographs indicated a relatively uniform dispersion of NZ nanoparticles at 2.5 and 5 wt% concentrations, with only a few random clusters. However, at 7.5 wt%, there was an increase in clustering and less uniform dispersion. Pin-on-disk results demonstrated reductions in the friction coefficient by up to 35 % and 47 %, and in the specific wear rate by up to 41 % and 54 % at 60 N and 100 N loads, respectively, with the inclusion of 5 wt% NZ. These improvements were attributed to optimal dispersion, reduced surface temperature, and increased hardness at the 5 wt% NZ level. The SVR-based regression model effectively captured the non-linear relationship between the input variables—CDI, load, and surface hardness—and the tribological responses (friction coefficient and specific wear rate). The results showed that loading magnitude and dispersion quality had the most significant effects on tribological performance, while surface hardness had the least influence. Morphological analysis of the worn surfaces revealed a reduction in sliding marks, abrasive tracks, ploughing signs, wear grooves, and adhesive wear with NZ inclusion, compared to neat PA6, at both 60 N and 100 N loads. These observations underscore the effectiveness of NZ in enhancing surface resistance to sliding frictional contact and thermal stresses during the pin-on-disk testing.

CRediT authorship contribution statement

R. Mohsenzadeh: Writing – review & editing, Writing – original draft, Supervision, Software, Project administration, Methodology,

Investigation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Statement on Originality

We confirm that this work is original and has not been published elsewhere, nor is it currently under consideration for publication elsewhere. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data Availability

Data will be made available on request.

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